

A PROJECT TITLE

ON

Water Quality Classification Using Logistic Regression and Decision Trees

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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DEPARTMENT OF CS & IT

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1. Abstract:

Access to safe and clean drinking water is fundamental to public health and sustainable development. However, increasing contamination risks demand robust, automated, and rapid methods for water quality assessment. This study investigates and compares the performance of two supervised machine learning algorithms, Logistic Regression and Decision Trees, for automated water potability classification.

Using a comprehensive dataset comprising key physicochemical parameters—such as pH levels, turbidity, hardness, solids, and chemical contaminants—the models were systematically trained, validated, and tested. Logistic Regression serves as an efficient, highly interpretable baseline well-suited for capturing linear relationships among water quality indicators. In contrast, the Decision Tree algorithm is deployed to model complex, non-linear interactions and threshold-based decision rules inherent in environmental data.

Model performance was evaluated using standard metrics, including accuracy, precision, recall, and the F1-score. Experimental findings indicate that while Logistic Regression offers computational efficiency and clear probabilistic insights, the Decision Tree model achieves superior overall classification performance by successfully capturing intricate feature interactions and non-linear patterns.

Ultimately, this research demonstrates the practical viability of deploying machine learning classifiers for real-time water quality monitoring. These automated approaches provide municipal authorities and water treatment facilities with reliable, scalable tools to streamline potability testing, mitigate public health hazards, and ensure compliance with safety standards.

2. INTRODUCTION:

Water is an indispensable resource for sustaining life, supporting ecosystems, and driving socioeconomic development worldwide. Despite its critical importance, securing safe and potable drinking water remains a formidable global challenge. Rapid industrialization, agricultural runoff, urbanization, and aging municipal infrastructure continuously threaten natural water bodies, introducing harmful pathogens, heavy metals, and chemical pollutants. Consuming contaminated water poses severe risks to human health, leading to waterborne diseases and long-term toxicological effects. Consequently, continuous and accurate water quality monitoring is essential for public health protection and regulatory compliance.

Traditionally, assessing water quality relies on manual sampling followed by laboratory-based physicochemical and microbiological analyses. While accurate, these conventional methods are labor-intensive, time-consuming, and expensive, often resulting in delayed detection of contamination events. In recent years, the integration of Artificial Intelligence (AI) and Machine Learning (ML) has revolutionized environmental monitoring, offering rapid, scalable,

and automated alternatives for predictive modeling and classification.

This study explores the application of supervised machine learning algorithms—specifically Logistic Regression and Decision Trees—to automate the classification of water potability. Logistic Regression provides a straightforward, highly interpretable probabilistic baseline for identifying linear relationships between water parameters and safety standards. Meanwhile, Decision Trees offer a flexible, non-parametric approach capable of capturing complex, hierarchical interactions and critical threshold limits among variables like pH, turbidity, and chemical concentrations. By evaluating and comparing the predictive performance of these two distinct algorithms, this research aims to determine the most effective computational framework for real-time water quality assessment. Ultimately, deploying these intelligent classification models can empower water treatment facilities and regulatory agencies to detect contamination swiftly, optimize resource allocation, and safeguard public health with greater efficiency.

3. SOFTWARE REQUIREMENTS:

- **Operating System:** Windows 10/11, Ubuntu 20.04+, or macOS (64-bit)
- **Language & IDEs:** Python 3.8+ using Jupyter Notebook, JupyterLab, or VS Code
- **Key Libraries:** Pandas, NumPy, Scikit-Learn, Matplotlib, and Seaborn
- **Hardware:** Dual-core CPU (quad-core recommended), 4–8 GB RAM, and 2 GB storage

Signature of the Guide

Course Instructor